

UEC752: IC FABRICATION TECHNOLOGY				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To gain knowledge about crystal growth and wafer preparation techniques. Subsequently, the thoroughly understanding of different integral steps needed for IC components fabrication mainly bipolar and field effect transistors. To acquire knowledge of various packaging techniques.				
Syllabus Integrated Circuits: Introduction, Impact of ICs on Industry, Advantages over discrete components, Monolithic and Hybrid ICs, Scales of integration and related issues. Crystalline and non-crystalline materials used in IC fabrication. Growth of Single Crystals wafers: Si crystal structure, Si crystal growth using Czochralski's method, Float Zone method and Bridgeman technique, characteristics and crystal evaluation, Wafer Shaping operations, Slicing, polishing and etching, Wafer cleaning. Epitaxy Film Formation: Importance of epitaxial layer growth, Types of epitaxy: VPE, MBE, MOCVD, Defects in epitaxial layers and their removal. Doping: Introduction, Basic concept of impurity diffusion in a semiconductor crystal. Fick's Laws, Analytical solution of diffusion equation, Gaussian solution near the surface and infinite medium. Ion-implantation, doping measurement methods. Subsequent Processes: Thermal Oxidation, Photolithography, Electron beam and X-Ray lithography, Positive and negative photo resist, dry and wet etching, Metallization using PVD, and clean room: Standards. MOSFET Technology: Design of junction diode, BJT, MOSFET and CMOS. Packaging of I.C's: Importance of packaging, Mountings in packages using Dual-in-line (DIP) or TO packages. Packages using surface-mount-technology (SMT). Familiarization with standards: IEEE 62659-2015.				
Laboratory Work:				
Course Learning Objectives (CLO) The students will be able to: 1. Explain crystal growth and wafer preparation techniques. 2. Understand different fabrication processes used in IC industry. 3. Apply IC fabrication technology to fabricate different semiconductor devices. 4. Explain various packaging techniques.				
Text Books 1. Sze, S. M., <i>VLSI Technology</i>, Wiley Eastern, USA (1999) 2nd ed. 2. J.D. Plummer, M.D. Deal and P.B. Griffin, <i>Silicon Vlsi Technology: Fundamentals</i>				

Practice and Modeling, Pearson Education Inc,(2009), 1st ed.

Reference Books

1. *Pucknell and Eshraghian, Basic VLSI Design, (2000) 2nd edition*
2. *Nagchoudhri, D., Principles of Microelectronics Technology (2002) 4th edition.*
3. *Sze, S. M., Semiconductor Devices, Physics & Technology, (2001) 3rd ed.*